FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

IT0049
(WEB SYSTEM TECHNOLOGIES)

TECHNICAL FORMATIVE ASSESSMENT

5

FROM ZERO TO FOUR PAGES, AGAIN: YOUR FIRST LARAVEL APPLICATION

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Members (if Group):	**Name** / **Role** (/) (/) (/) (/)
Section:	TC37
Professor:	Mr. Ric Cristian Frias

PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Analyze a complex problem and identify and define the computing requirements appropriate to its solution. [PO: B]

COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- Understand the advanced principles of web development using web frameworks and MVC architecture. [CLO: 1]

INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Set up a new Laravel project and run it locally.
- Use Artisan to inspect and generate application files.
- Create Laravel routes, controllers, and Blade views.
- Use a static PHP array as a temporary data source for a listing page.

BACKGROUND INFORMATION

This activity replicates TFA1, but in Laravel. The MVC responsibilities are exactly the same as CodeIgniter's — a route decides which controller method runs, the controller decides what happens, and a view (here, a Blade template) renders the response — only the file names, folder structure, and syntax differ. Focus on recognizing the same responsibilities rather than learning the ideas from scratch.

GRADING SYSTEM / RUBRIC

CRITERIA	Description
Functionality & Requirements Completeness (40 points max)	All required pages and features are present and work as specified in the activity instructions.
Code Structure & Organization (25 points max)	Controllers, models, and views are separated correctly and follow the framework's conventions; code is readable and sensibly named.
Data Handling with Static Arrays (20 points max)	The static PHP array is structured sensibly, used consistently as the page's temporary data source, and looped through correctly in the Blade view.

CRITERIA	Description
Documentation & Submission Quality (15 points max)	The GitHub repository is organized and complete, the README explains how to set up and run the project, and the hosted link is live and matches the submitted code.

LABORATORY ACTIVITY

Rebuild the POS application from TFA1 in Laravel: a landing page (/), an about page (/about), a Customer Accounts page (/customers), and a User Accounts page (/users), with the account pages using a static PHP array as a temporary data source.

1. Create a new Laravel project and confirm it runs locally with the Artisan development server.
2. Define routes for all four pages in routes/web.php.
3. Create a PageController for the landing and about pages, and a CustomerController and UserController for the account pages, each defining a static PHP array of at least 5 sample records.
4. Create Blade views for all four pages, using @foreach to loop through the static array data in the Customer Accounts and User Accounts views.
5. Add navigation links between all four pages and confirm every route works.

Submission Requirements:

- A link to a GitHub repository containing your raw project files, including your database export or migration and seeder files.
- A link to your hosted, working version of the application.

REFERENCES

- Stauffer, M. (2024). Laravel: Up & Running: A Framework for Building Modern PHP Apps (3rd ed.).
- Laravel Documentation. Routing. https://laravel.com/docs/13.x/routing
- Laravel Documentation. Blade Templates. https://laravel.com/docs/13.x/blade

1. Create a New Laravel Project and Confirm It Runs Locally with the Artisan Development Server

I created a new Laravel project named **LARAVEL** inside my IT0049 – Web System Technologies directory using the Laravel installer. After completing the installation and configuring the initial setup options, I opened the project folder in the VS Code terminal.

To verify the installation, I executed the command php artisan --version, which confirmed that the project was running **Laravel Framework 12.70.2**. I then started the local development server using php artisan serve.

The terminal confirmed that the Laravel development server was running at http://127.0.0.1:8000. I opened this address in Google Chrome and successfully accessed the default Laravel welcome page.

This confirms that the Laravel project was installed successfully and that the local development environment is working. The project is now ready for the implementation of the required POS pages, routes, controllers, and Blade views.

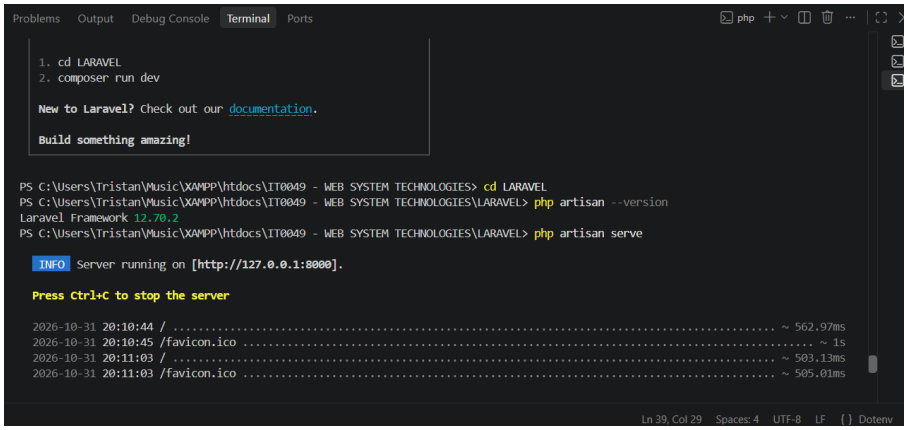


Figure 1. Laravel framework version and initial server. Proves Laravel was installed and runs locally

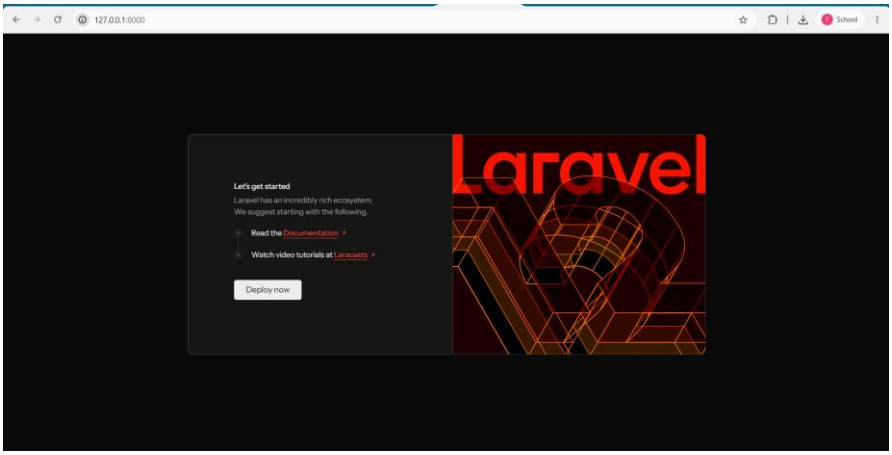


Figure 2. Laravel welcome page in Chrome. Proves the original Laravel installation works

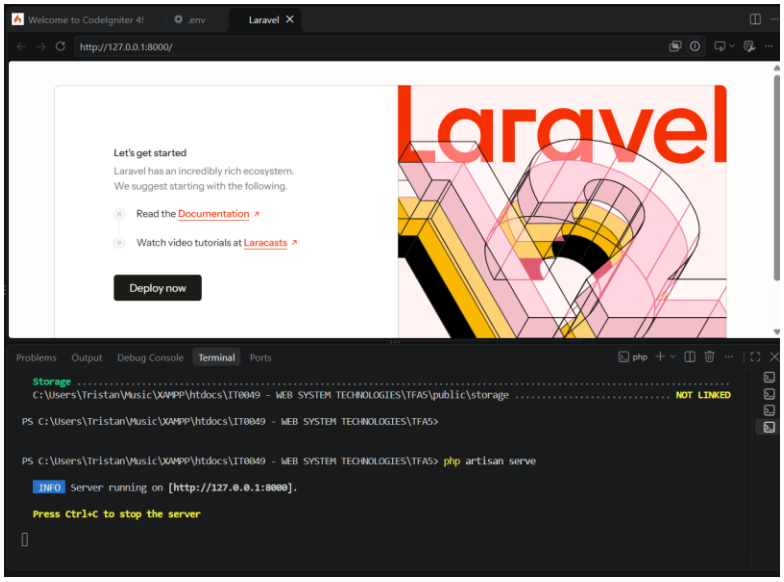


Figure 3. TFA5 server and welcome page. Proves the assessment has its own working Laravel project

Figure 3. Verification of the TFA5 Laravel Project

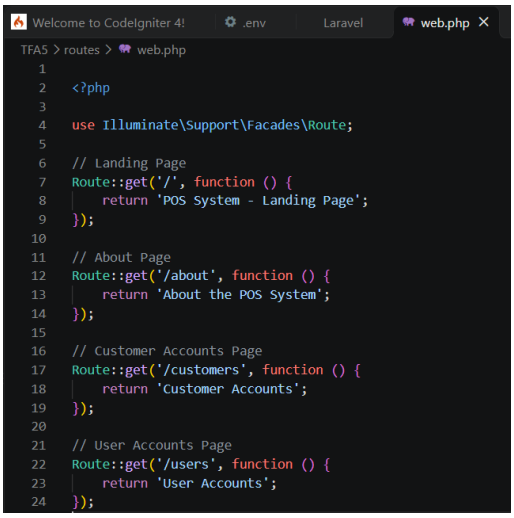
After successfully installing Laravel, I created a separate project folder named TFA5 based on my configured Laravel installation. This allows me to develop the assessment independently without modifying the original Laravel project.

I configured the TFA5 application's environment settings and generated a new application key using php artisan key:generate. I then verified the project configuration using php artisan about and started its local development server using php artisan serve.

The application successfully opened at http://127.0.0.1:8000/, displaying the default Laravel welcome page. This confirms that the TFA5 project is functioning locally and is ready for the implementation of the required POS pages.

2. Define Routes for All Four Pages

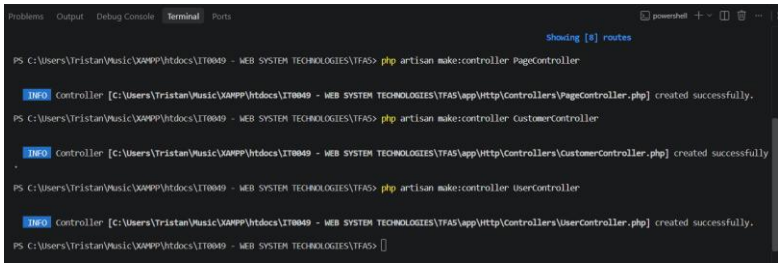
I configured routes/web.php to register the four required POS routes: /, /about, /customers, and /users. I verified the routes using php artisan route:list and confirmed that each URL responds successfully. The routes will be connected to their respective controllers in the next step.

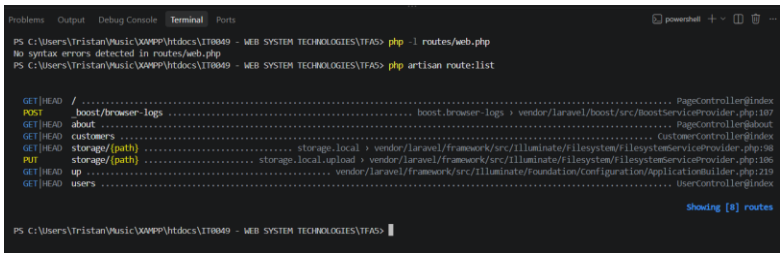




3. Create the Required POS Controllers

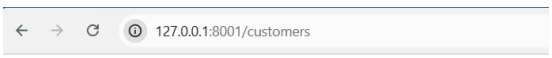
I created three Laravel controllers using Artisan: PageController, CustomerController, and UserController. PageController handles the landing and about pages, while the other two controllers contain five static PHP records each for customer and staff accounts. I connected the four routes to their corresponding controller methods and verified the PHP syntax and route registrations.





4. Create Blade Views for All Four Pages

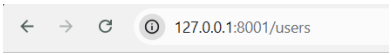
I created four Blade views for the Landing, About, Customer Accounts, and User Accounts pages. The account pages use @foreach to display five sample records from their respective controllers. I tested each page locally and confirmed that the information displays correctly.



Customer Accounts

List of registered customers in the POS system.

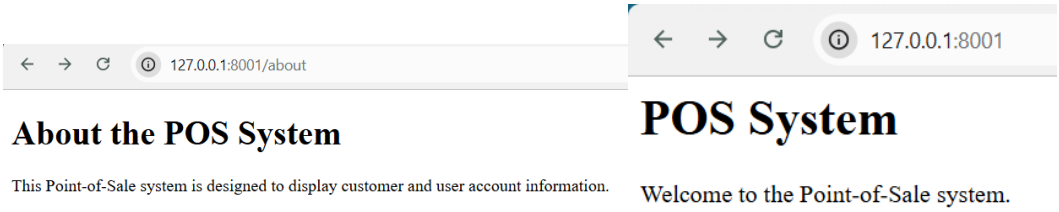
Full Name	Email	Phone Number
Juan Dela Cruz	juan.delacruz@example.com	09171234567
Maria Santos	maria.santos@example.com	09182345678
Carlo Reyes	carlo.reyes@example.com	09193456789
Angela Garcia	angela.garcia@example.com	09204567890
Mark Mendoza	mark.mendoza@example.com	09215678901



User Accounts

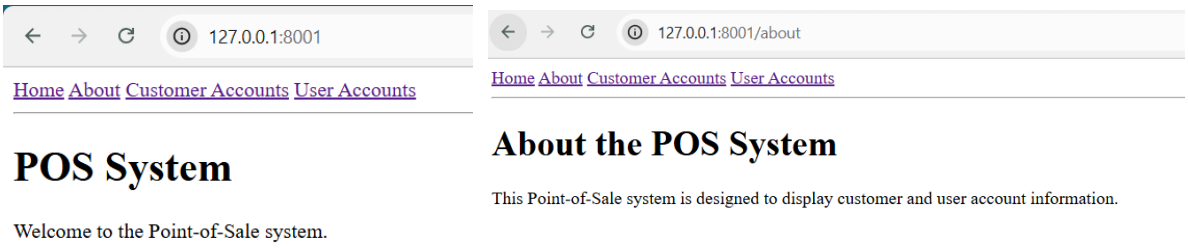
List of staff accounts in the POS system.

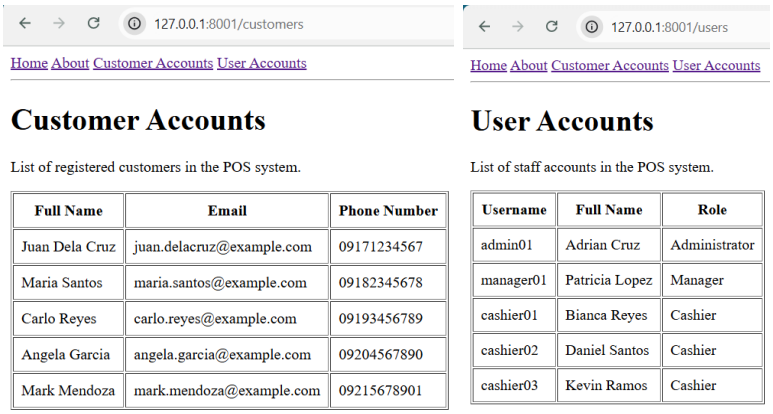
Username	Full Name	Role
admin01	Adrian Cruz	Administrator
manager01	Patricia Lopez	Manager
cashier01	Bianca Reyes	Cashier
cashier02	Daniel Santos	Cashier
cashier03	Kevin Ramos	Cashier



5. Add Navigation Links Between All Four Pages

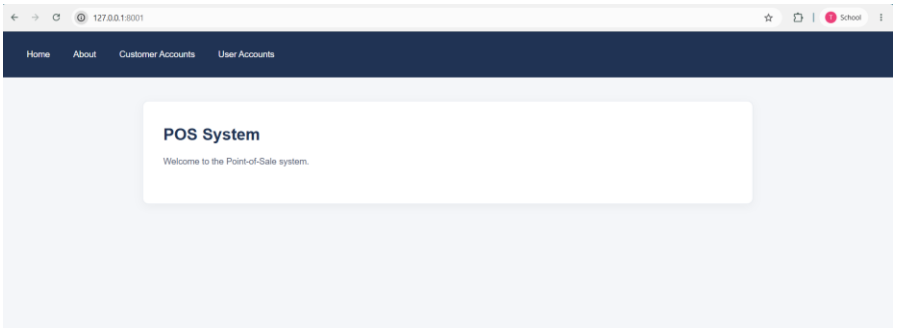
I created a shared Blade layout containing navigation links for Home, About, Customer Accounts, and User Accounts. All four pages use the same layout, allowing users to navigate between them without duplicating the navigation code.

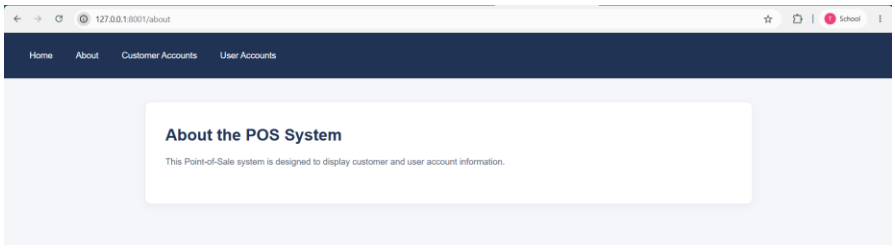


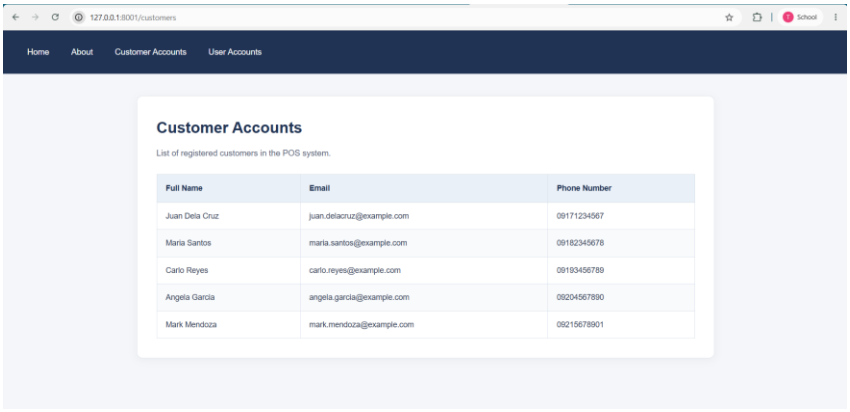


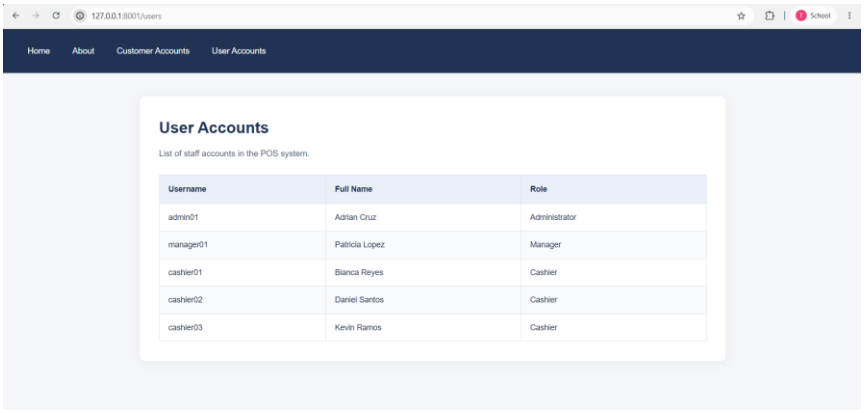
5. Add Navigation Links Between All Four Pages

I created a shared Blade layout that provides navigation between the Home, About, Customer Accounts, and User Accounts pages. I also added an external CSS stylesheet to improve the appearance of all four pages without changing their content or functionality. The shared layout keeps the navigation and design consistent throughout the POS system.


127.0.0.1:8001
School
Home
About
Customer Accounts
User Accounts
POS System
Welcome to the Point-of-Sale system.


127.0.0.1:8001/about
School
Home
About
Customer Accounts
User Accounts
About the POS System
This Point-of-Sale system is designed to display customer and user account information.


127.0.0.1:8001/customers
School
Home
About
Customer Accounts
User Accounts
Customer Accounts
List of registered customers in the POS system.
Full Name
Email
Phone Number
Juan Dela Cruz
juan.delacruz@example.com
09171234567
Maria Santos
maria.santos@example.com
09182345678
Carlo Reyes
carlo.reyes@example.com
09193456789
Angela Garcia
angela.garcia@example.com
09204567890
Mark Mendoza
mark.mendoza@example.com
09215678901


127.0.0.1:8001/users
School
Home
About
Customer Accounts
User Accounts
User Accounts
List of staff accounts in the POS system.
Username
Full Name
Role
admin01
Adrian Cruz
Administrator
manager01
Patricia Lopez
Manager
cashier01
Bianca Reyes
Cashier
cashier02
Daniel Santos
Cashier
cashier03
Kevin Ramos
Cashier